WORKED SOLUTION

# STEP I 1993, Question 8

Pure • Spheres and Vectors

**STEP I 1993, Question 8**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q8

## The key idea

For part (i), put the origin and axes where the symmetry is. If the centre of the sphere sits on the $z$-axis and the plane is $z = 0$, the equations do almost all the work. Two spheres are then no harder, because subtracting their equations cancels every square and leaves the equation of a *plane*. In part (ii), coordinates again reduce everything to a scalar product. Put $B$ at the origin, so that both pole tops are measured from $E$.

## Solution

### Part (i)

**Sphere and plane.** Let the sphere have centre $O'$ and radius $r > 0$, and let $N$ be the foot of the perpendicular from $O'$ to the plane. Take $N$ as origin, the plane as $z = 0$, and the $z$-axis along $NO'$. Then $O' = (0, 0, d)$, where $d \geq 0$ is the distance from the centre to the plane. The sphere is

$$x^2 + y^2 + (z - d)^2 = r^2,$$

so a point lies on both the sphere and the plane exactly when $z = 0$ and

$$x^2 + y^2 = r^2 - d^2.$$

- If $d < r$, this is the circle in the plane with centre $N$ and radius $\sqrt{r^2 - d^2}$.
- If $d = r$, then $x^2 + y^2 = 0$ forces $x = y = 0$, so the intersection is the single point $N$ (the plane is a tangent plane).
- If $d > r$, the right-hand side is negative, so there are no points and the intersection is empty.

These cases cover every possibility, which proves the first statement.

**Two spheres.** Let the spheres have radii $r, s > 0$ and distinct centres a distance $c > 0$

apart. Put the first centre at the origin and the second at $(0,0,c)$:

$$x^2+y^2+z^2=r^2, \qquad x^2+y^2+(z-c)^2=s^2.$$

**Ask yourself** How can two quadratic equations have a solution set that is just a circle?

Subtract the second equation from the first. All the squared terms cancel, leaving

$$2cz-c^2=r^2-s^2 \iff z=h, \qquad \text{where } h=\frac{r^2-s^2+c^2}{2c}.$$

Dividing by $c$ is allowed because the centres are distinct. A point lies on both spheres if and only if it lies on the first sphere and satisfies the difference equation, so

$$(\text{sphere } 1)\cap(\text{sphere } 2)=(\text{sphere } 1)\cap(\text{plane } z=h).$$

By the first result this is a circle, a point or the empty set. (Explicitly, it is the circle $x^2+y^2=r^2-h^2$ in the plane $z=h$ when $|h|<r$, a single point when $|h|=r$, and empty when $|h|>r$.) The circle lies in a plane perpendicular to the line of centres, and its centre is on that line.

### Part (ii)

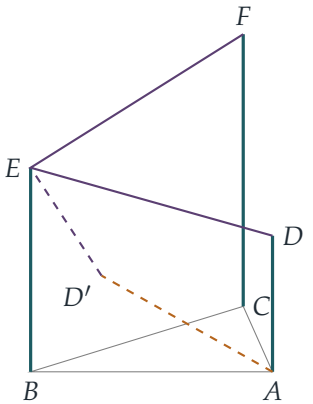
**Assumption about the bases.** The question does not say where $A$, $B$ and $C$ stand. We take them to be at the vertices of an equilateral triangle of side 2 m on level ground. This is the configuration that gives the stated angle (see the Checks).

Take $B$ as origin with the $z$-axis vertical:

$$B=(0,0,0), \quad A=(2,0,0), \quad C=(1,\sqrt{3},0),$$

so that the tops of the poles are

$$D=(2,0,2), \quad E=(0,0,3), \quad F=(1,\sqrt{3},4).$$



In the diagram, $D'$ (dashed) is the position of $D$ after the vandals have moved the pole.

**The original sculpture.** Using vectors from $E$,

$$\overrightarrow{ED} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \qquad \overrightarrow{EF} = \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix},$$

so $\overrightarrow{ED} \cdot \overrightarrow{EF} = 2 + 0 - 1 = 1$ and $|\overrightarrow{ED}| = |\overrightarrow{EF}| = \sqrt{5}$. Therefore

$$\cos \angle DEF = \frac{1}{\sqrt{5}\sqrt{5}} = \frac{1}{5}, \qquad \text{so} \qquad \angle DEF = \cos^{-1} \tfrac{1}{5},$$

as required.

**After the vandals.** The pole $AD$ still has length 2 and still stands at $A$. It now lies in the vertical plane through $A$ and $B$ (the plane $y = 0$), leaning towards $B$ at $\frac{1}{4}\pi$ to the ground. Moving from $A$ towards $B$ is the direction of $-x$, so

$$D' = \left(2 - 2\cos \tfrac{\pi}{4},\ 0,\ 2\sin \tfrac{\pi}{4}\right) = \left(2 - \sqrt{2},\ 0,\ \sqrt{2}\right).$$

Since $2 - \sqrt{2} \approx 0.59$ lies between 0 and 2, the point $D'$ really is above the segment $AB$. Now

$$\overrightarrow{ED'} = \begin{pmatrix} 2 - \sqrt{2} \\ 0 \\ \sqrt{2} - 3 \end{pmatrix}, \qquad \overrightarrow{ED'} \cdot \overrightarrow{EF} = (2 - \sqrt{2}) + (\sqrt{2} - 3) = -1,$$

and

$$|\overrightarrow{ED'}|^2 = (2 - \sqrt{2})^2 + (3 - \sqrt{2})^2 = (6 - 4\sqrt{2}) + (11 - 6\sqrt{2}) = 17 - 10\sqrt{2}.$$

**Ask yourself** The dot product is negative. What does that tell us about the angle?

A negative scalar product means the angle is obtuse, so we need the value of $\cos^{-1}$ that lies between $\frac{1}{2}\pi$ and $\pi$:

$$\cos \angle D'EF = \frac{-1}{\sqrt{5}\sqrt{17 - 10\sqrt{2}}} = \frac{-1}{\sqrt{85 - 50\sqrt{2}}} \approx -0.264542,$$

$$\angle D'EF \approx 1.839 \text{ radians (to four figures).}$$

## Checks

- With an equilateral base of side $s$, the same method gives $\cos \angle DEF = \dfrac{s^2/2 - 1}{s^2 + 1}$. This equals $\frac{1}{5}$ only when $s = 2$, which supports the assumed base. A numerical check confirms $\cos \angle DEF = 0.2$ for $s = 2$.
- $|AD'| = 2$ and $\angle BAD' = \frac{1}{4}\pi$ were checked numerically. The new angle came out as 1.838525… radians (about 105.3°), agreeing with the exact form above.
- Part (i): random spheres were tested with points on the predicted circles; they lay on both surfaces to within $10^{-12}$.

## Common slips

- In the two-sphere case, claiming that the difference of the equations *is* the intersection. The plane only contains it; we still have to intersect the plane with one of the spheres.
- Placing $D'$ on the far side of $A$, away from $B$. That point is not above the segment $AB$, and it gives a different angle.
- Giving the answer in degrees, or as the acute angle $\cos^{-1}(0.2645)$, after losing the minus sign.

## Beyond STEP

Subtracting the equations of two spheres to get a plane is the idea behind the *radical plane* (in two dimensions, the radical axis) of two circles or spheres. The same trick turns GPS-style trilateration into a linear problem. More generally, the intersection of any quadric surface with a plane is a conic, and classifying such intersections is a good first exercise in linear algebra with quadratic forms. Strang's *Introduction to Linear Algebra* develops the tools, and Siklos's *Advanced Problems in Mathematics* has more STEP-style coordinate geometry.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.